

ANTHONY CATALINA

Frontend Engineer · React & TypeScript · Real-Time UI & Design Systems

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SUMMARY

Frontend engineer with 5+ years of experience delivering production React/TypeScript applications at enterprise scale. Unique background spanning both embedded systems and modern UI development; enabling end-to-end ownership from hardware data pipelines through polished, performant interfaces. Strong track record building real-time dashboards, component libraries, and design systems for Fortune 500 clients across IoT, industrial, and distribution industries.

SKILLS

Frameworks & Languages: React, TypeScript, JavaScript, Lit, HTML5, CSS3/SCSS, Bootstrap, Python

Architecture & Patterns: Atomic Design, Component-Driven Development, MVC, Monorepo, Design Systems, REST APIs

Real-Time Data: MQTT, WebSocket, Live data pipelines

DevOps & Tooling: AWS S3, GitHub Actions CI/CD, Jenkins, Git, Bitbucket, SureMDM

EXPERIENCE

Embedded UI Engineer · **Hach Company — Loveland, CO** · Hybrid

Jan 2026 – Present

Contract Frontend Engineer

Apr 2025 – Jan 2026

Promoted from Contract role; bridging embedded systems and frontend UI across multiple enterprise product lines

- Lead frontend architecture for a large-scale TypeScript/Lit monorepo supporting multiple water analysis product lines; established reusable Atomic UI component libraries and MVC patterns adopted across concurrent product teams.
- Engineer real-time device management interfaces integrating live MQTT and HTTP data streams, delivering low-latency UI updates critical for lab and field instrumentation workflows.
- Define and enforce frontend engineering standards through custom ESLint rule sets and comprehensive Jest unit test suites; manage UI submodule dependencies across product lines to ensure stable, coordinated releases.
- Act as the technical bridge between embedded firmware teams and UI — translating hardware data pipelines into intuitive, responsive interfaces for engineers, lab technicians, and enterprise clients.

Stack: TypeScript · Lit · React · MQTT · Embedded C/C++ · Jest · ESLint · Jenkins · Bitbucket

Software Engineer · **Philadelphia Scientific, LLC — Montgomeryville, PA**

Sept 2022 – Apr 2025

- Led full-stack frontend development of multiple production web applications for Fortune 500 clients including Target, TJX, and Uline — independently owning the full lifecycle from component design through AWS deployment.
- Architected a real-time operations dashboard with interactive data visualizations and live data feeds; built zero-downtime CI/CD pipeline via GitHub Actions with automated deployment to AWS S3.
- Built a touch-optimized React PWA for on-device battery maintenance with a Python/Flask REST API backend; deployed and managed enterprise-wide across major national distribution centers via SureMDM MDM.
- Designed React/TypeScript component libraries with Bootstrap, enforcing consistent UX patterns and accessibility across all client-facing products; provided ongoing production support maintaining uptime for business-critical applications.

Stack: React · TypeScript · Python · Flask · SQL · AWS S3 · Bootstrap · GitHub Actions · SureMDM

Software Engineer · **Regulus Group, LLC — Atlantic City, NJ**

Jun 2021 – Sept 2022

- Modernized data-input and analytical-output UIs for FAA Surveillance Branch operators, upgrading legacy interfaces with improved usability and performance for mission-critical government systems.
- Developed real-time and historical system performance dashboards with dynamic data visualizations; optimized SQL queries for high-volume, data-intensive analytical workloads.

Stack: JavaScript · HTML/CSS · SQL · Dashboard Development

PROJECTS

Water Analysis Instrument UI · **Hach**

- Production UI serving engineers, lab technicians, and enterprise clients across multiple water analysis product lines; fully modular Atomic UI component system built for reuse and long-term maintainability.

Stack: TypeScript · Lit · React · MQTT · Jest · Atomic Design

Battery Manager Operations Dashboard · **Philadelphia Scientific**

- Enterprise real-time ops dashboard with live data visualization and interactive charts; fully automated AWS S3 + GitHub Actions deployment pipeline serving Fortune 500 distribution center operators.

Stack: React · TypeScript · Python · SQL · AWS S3 · GitHub Actions

EDUCATION

B.S. Computer Science · Rowan University, Glassboro, NJ

December 2020